

Justify your answers and show your work.

Find the limit of the following sequences or determine that the limit does not exist.

9.2:9. $\left\{ \frac{n^3}{n^4+1} \right\}$

9.2:43. $\left\{ \frac{\sin n}{2^n} \right\}$

see **9.2:51-56.** $\left\{ \frac{n^2}{n^2 \ln n} \right\}$

Evaluate the geometric series or explain why it diverges.

9.3:19 & 23. $\sum_{k=0}^{\infty} \left(\frac{1}{4} \right)^k$ and $\sum_{k=0}^{\infty} (1.01)^k$

Consider the following infinite series.

9.1:55. $\sum_{k=1}^{\infty} \frac{1}{4k^2 - 1}.$

a. Write out the first four terms of the partial sums.

b. Use the results of part (a) to propose an explicit formula for S_n .

bonus. Prove that $\sum_{k=1}^{\infty} 2k - 1 = n^2.$